

INTRODUCTION TO CIVIL ENGINEERING



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CIVIL ENGINEERING: AN OVERVIEW

- Civil Engineering is the engineering discipline that encompasses planning, designing and constructing the built environment for human comfort, safety and sustainability of life. Civil engineering is among the oldest engineering disciplines, with roots stretching back to the beginnings of human civilization.
- Humans adapted cave shelters to protect themselves of extreme weather and harsh environment, and used tree trunks to cross rivers, demonstrating the ancient age of civil engineering. It is considered as the second engineering discipline historically.
- The first type of engineering is considered to be a profession that was military engineering. The new field was a response to this and was a way for engineers working on non-military projects to set themselves apart from military engineering
- Prof. Ravindra Gettu What is civil engineering ?, Lecture 1, Introduction to Civil Engineering - CE1010, Course 2 - Second trimester (Jan-Mar) 2021, Coordinators: Prof. Ravindra Gettu and Dr. Ramesh Kannan K
- An Introduction To Civil Geoengineering://www.bioengineering.com/resources/what-is-civil-engineering
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CIVIL ENGINEERING: AN OVERVIEW (https://webpages.charlotte.edu/~jmconrad/hsed/cveg_0in.pdf)

- Civil engineering affects many of our daily activities: the buildings we live in and work in, the transportation facilities we use, the water we drink, and the drainage and sewage systems that are necessary to our health and well-being. Civil engineers:
- Measure and map the earth's surface.
- Design and supervise the construction of bridges, tunnels, large buildings, dams, and coastal structures.
- Plan, layout, construct, and maintain railroads, highways, and airports.
- Devise systems for the control and efficient flow of traffic.
- Plan and build river navigation and flood control projects.
- Provide plants and systems for water supply and sewage and refuse disposal.

CIVIL ENGINEERS ROLE (https://webpages.charlotte.edu/~jmconrad/hsed/cveg_0in.pdf)

- Civil engineering is much more than erecting skyscrapers or bridges. Civil engineers are trained in the interactions among structures, the earth, and water, with applications ranging from highways to dams and water reservoirs. Deeply involved with specifying appropriate construction materials, many civil engineers and others are also employed by the manufacturers of those materials. Since constructing a large building or public-works project can involve elaborate planning, civil engineers can be outstanding project managers. They sometimes oversee 0thousands of workers and develop advanced computerization and planning policies.
- Most significantly, many civil engineers are involved with preserving, protecting, or restoring the environment. Most water treatment and water purification projects are designed and constructed by civil engineers (in these two areas, many of them are known as environmental engineers). A growing number of civil engineers are involved in billion-dollar projects to clean up toxic industrial or municipal wastes at abandoned dump sites. Civil engineers engage in such diverse projects as preserving wetlands or beaches, maintaining national forest parks, and restoring the land around mines, oil wells, or factories.

CIVIL ENGINEERS ROLE (https://webpages.charlotte.edu/~jmconrad/hsed/cveg_0in.pdf)

- The key theme in the civil engineering Canon of Ethics is the “protection of the welfare and safety of the public”. Indeed, the “civil” in civil engineering refers to the discipline’s involvement in public works, including government buildings, military bases, water treatment works, mass transit systems, airports, shipping ports, and parks. Because of this involvement, many civil engineers find themselves employees of, or suppliers for, local government. This relationship, combined with the requirements for public safety, translates into a high degree of professionalism. A key word that arose in the 1980s and will remain important for civil engineers for many years to come is “infrastructure”. Infrastructure demands will be a big element of civil engineering for the next several years.
- Civil Engineers have a wide range of career options from which to choose. Civil engineers work with construction companies, manufacturing companies, power companies, and with consulting engineering firms. Many opportunities for civil engineering employment exist in city, county, state, and the government agencies.

CIVIL ENGINEERS ROLE

- Civil Engineers shape the built environment, and use their creativity to improve the world we live in for a sustainable future. Every building, road, bridge, dam, canal, harbour, airport or energy facility will have relied on the creativity, imagination and skills of Civil Engineers.
- However, in the modern world these existing projects have aged and civil engineers are now also needed to manage construction projects to rebuild, repair, and upgrade bridges, roads, levees, dams, airports, buildings, and structures of all types, making it an in-demand and highly lucrative field.
- Today Civil engineers design, construct, maintain, and operate infrastructures, while simultaneously safeguarding public and environmental health and upgrading neglected structures and systems.

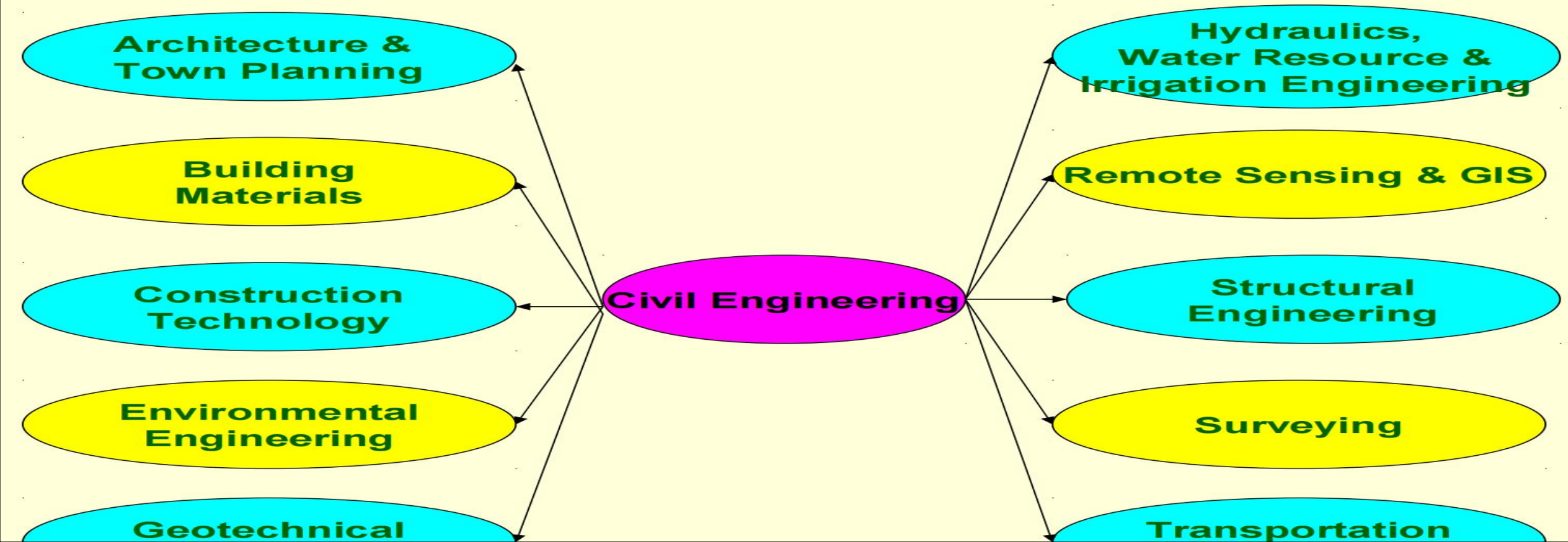
Understanding the Scope of Civil Engineering

<https://projectinfrastructure.com/civil-engineering/>

; <https://krishnakumar31.wordpress.com/wp-content/uploads/2017/09/civilintroduction-130920022924-phpapp01.pdf>

- Civil Engineering is an umbrella term that covers multiple specialized branches. While the field is incredibly diverse, many of the sub-disciplines intersect and collaborate in real-world projects. Understanding these specializations helps provide a fuller picture of the civil engineering profession.

DISCIPLINES OF CIVIL ENGINEERING



Structural Engineering

- Structural engineers analyze and design the 'skeleton' or framework of buildings, bridges, tunnels, and other large infrastructure components. They ensure that these structures can withstand internal and external forces, such as load, wind, earthquakes, and vibrations. Structural engineering often involves the use of advanced materials, computational modeling, and strict safety standards. The main component is the analysis and design of structures to resist loads and other actions, and to ensure their functionality and serviceability. Considerations of economics, aesthetics and social implications is also a necessity. The base of structural engineering is:
 - Mathematics (for modelling the stimulus and its response, numerical methods)
 - Applied Mechanics and Strength of Materials
 - Structural analysis (for obtaining deformations and stresses)
 - Structural design (for providing safe and usable structures)

Structural Engineering (<https://krishnakumar31.wordpress.com/wp-content/uploads/2017/09/civilintroduction-130920022924-phpapp01.pdf>)



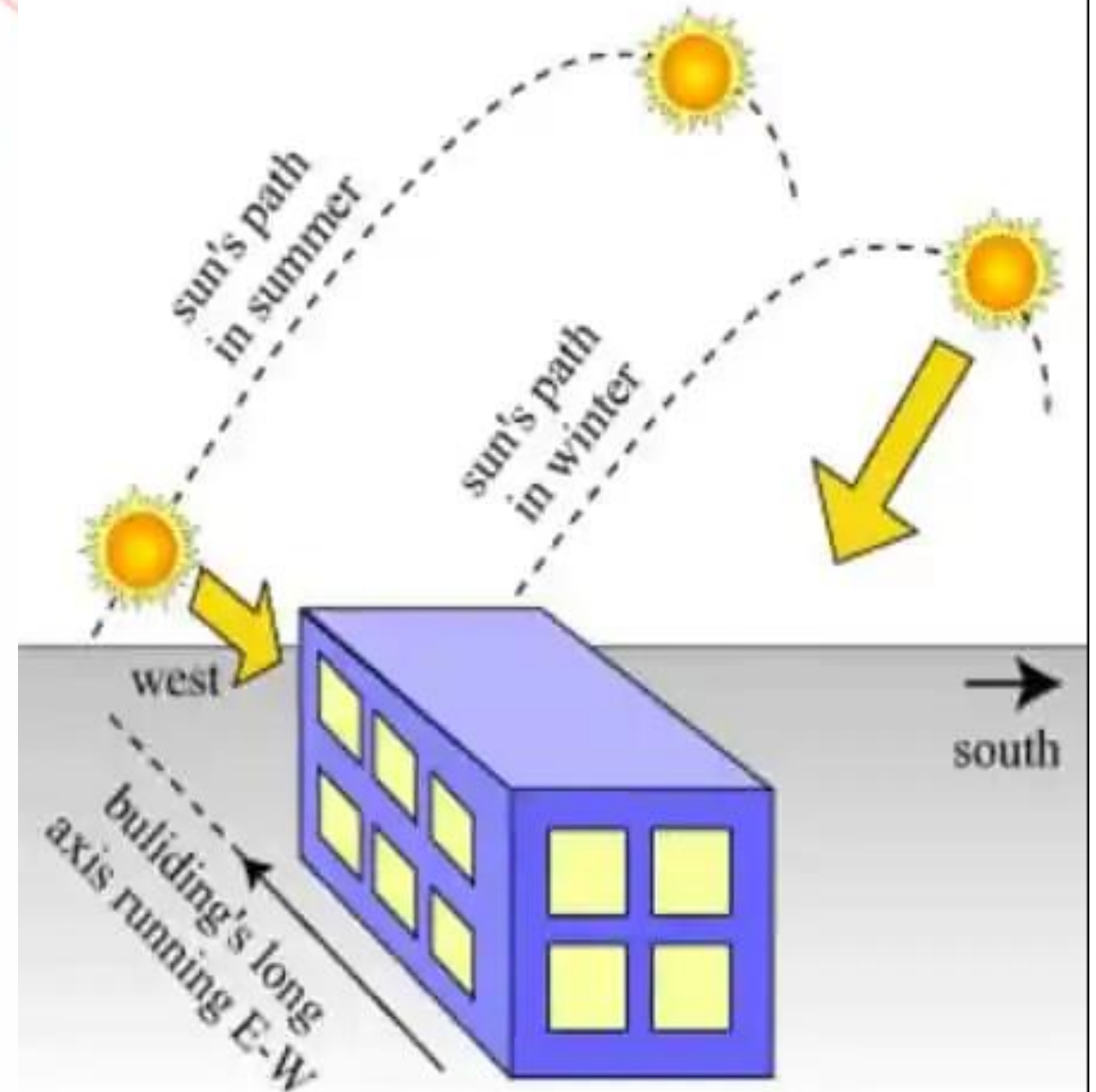
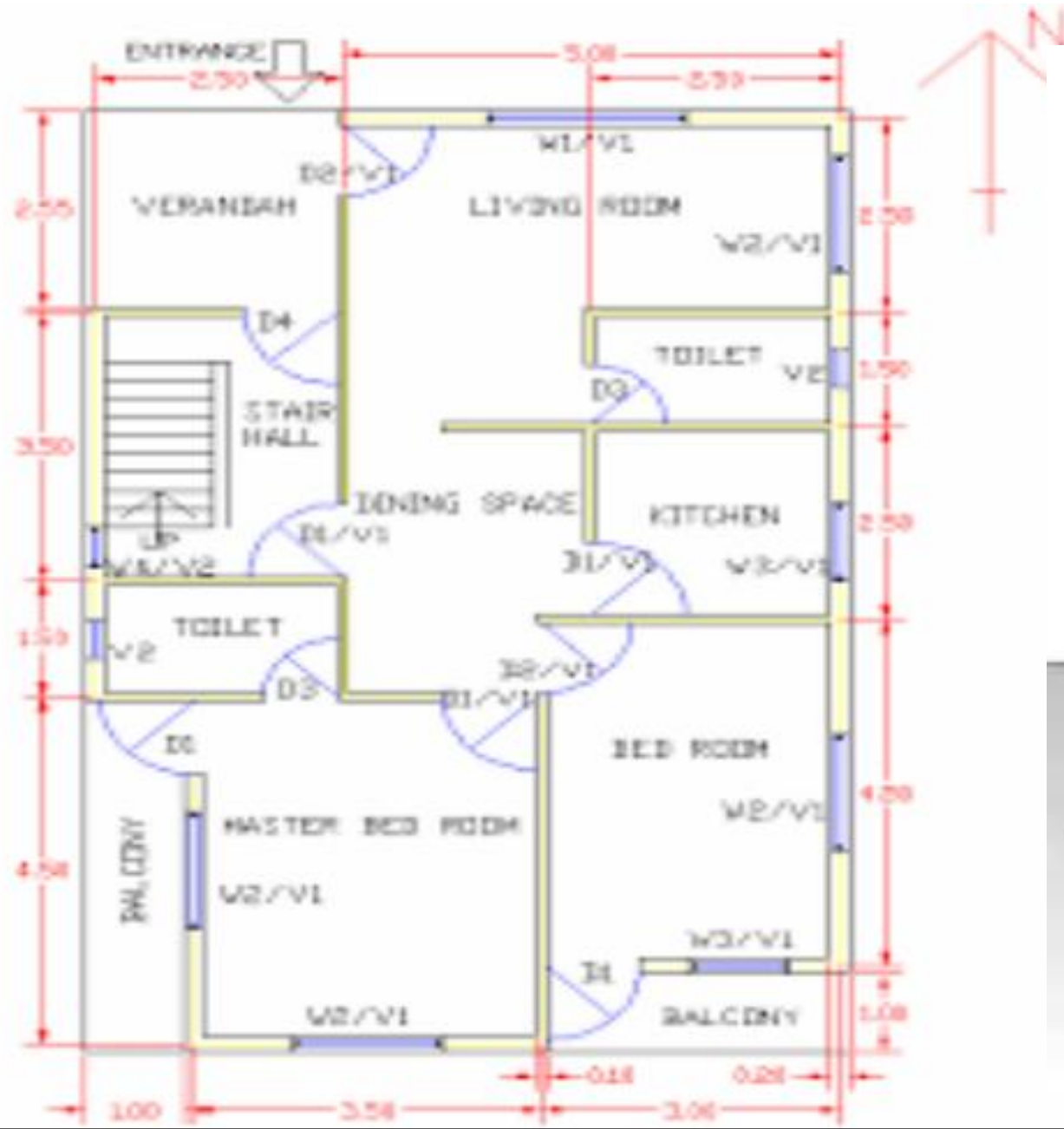
Building Planning and Construction engineering

- Civil engineers have often been associated with building services, working as a structural engineer within the design process to produce solutions to architectural problems and providing specialist understanding of modern construction methods. These engineers work alongside those in Architectural Engineering to ensure buildings get designed safely and meet the requirements of evolving government regulations. Moreover they are in-charge of the building construction and its project management.
- In recent times Building Information Modeling (BIM) has also propped which deals with the operation of the systems in large buildings which is mostly handled by the Civil Engineers.
- Today engineers have ethical and professional responsibilities to ensure that buildings are developed with the climate-crisis in mind. Sustainable engineering materials and building techniques are an important part of how engineers can combat climate change.

Building Planning and Construction engineering

- Building planning is the method of setting or arranging various components or units of a building in a systematic manner to form a meaningful and homogeneous structure to meet its functional purpose. Building Planning depends on its functional use like Residential buildings, Educational buildings, Institutional buildings, Assembly buildings, Public buildings, Business buildings, Mercantile buildings, commercial buildings, industrial buildings, Storage buildings, Hazardous buildings, etc.
- Buildings nowadays are not only structures like foundations, floors, beams, columns, slabs walls, staircases and doors and windows but they are heading towards smart buildings in which the control on occupation and movement within is becoming more automatic and ICT based.

Building Planning and Construction engineering



Building Construction

<https://www.aboutcivil.org/building-construction-and-design.html>



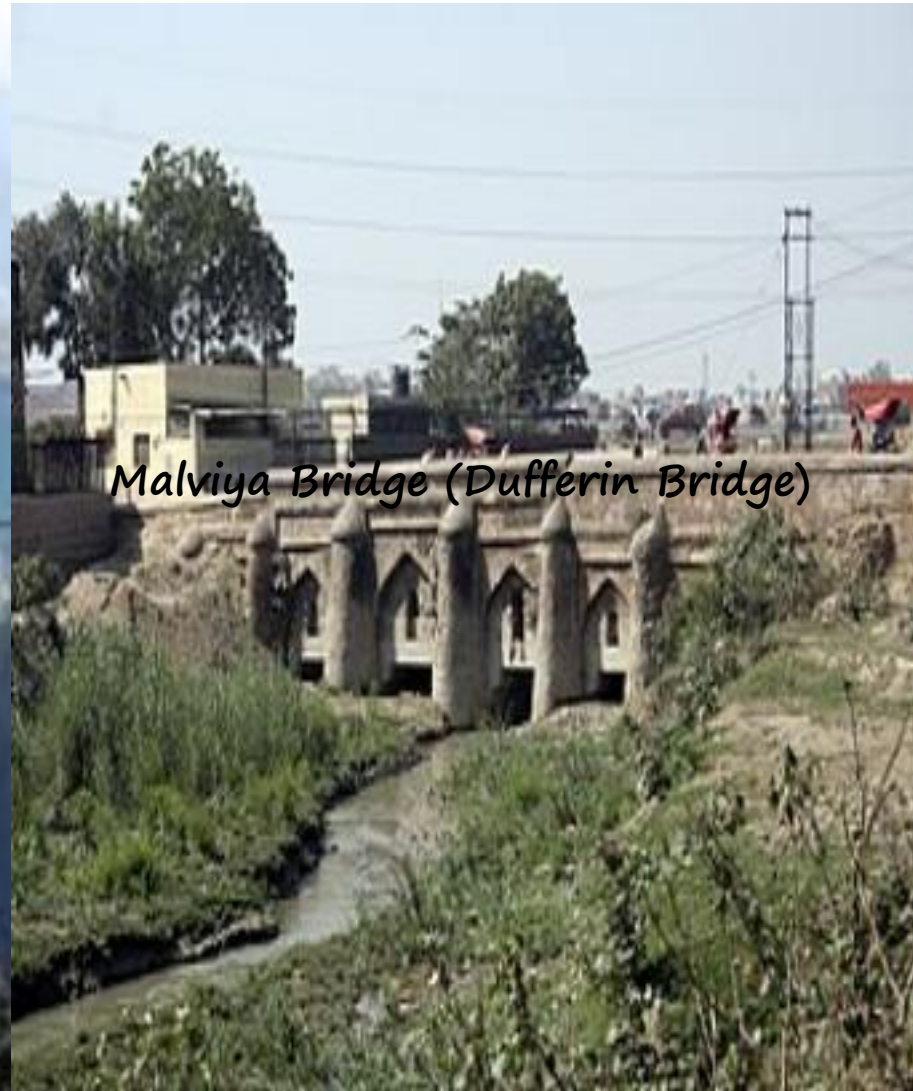
Bridge Engineering

- Bridge engineering is one of the most associated field with big Civil Engineering projects but it also requires several sub-specializations to work together. For example, Engineers working on the design and construction management of a bridge need a strong understanding of Structural Engineering and sustainable Principles but they also need to be able to work alongside road, geotechnical, railway and environmental engineers. If you are to function effectively as a member or leader of a team that establishes goals, plan tasks, meets deadlines, and creates a collaborative and inclusive environment is as important to your career as your technical knowledge. Bridges usually are constructed over the water bodies to join two banks or ends. However there are various other kinds of bridges also viz. the Level Separators or Viaducts. The Viaduct is a specific type of bridge designed to carry a road or railway over long distances, typically across valleys, urban areas, or uneven terrain. Unlike general bridges, viaducts consist of a series of uniform spans supported by regularly spaced piers or columns. (<https://www.encardio.com/blog/urban-transportation-viaducts-vs-bridges>)
- Grade separated intersection is a bridge that eliminates crossing conflicts at intersections by vertical separation of roadways in space. Grade separated intersection are otherwise known as Interchanges. Various types of structures are possible in a grade separated intersection. (https://www.civil.iitb.ac.in/tvm/nptel/567_Grade/web/web.html)

Bridge Engineering



Chenab Rail Bridge in 2023



Malviya Bridge (Dufferin Bridge)

Wazirabad Bridge (14th century)



Malviya Bridge (Dufferin Bridge)

Via Duct and Grade Separated Intersection



Hydraulic Civil Engineering (Hydraulics)

It deals with the application of fluid mechanics to water flowing in any isolated environment (for example a pipe or a pump) or how water moves in an open channel (river, lake, ocean, canals) and often involves engineers being asked to solve complex engineering problems. A few examples of important components of hydraulics are

1. Planning and Design of hydraulic structures like dams, weirs and Barrages for storage and diversion of water in the rivers.
2. Hydro-power Production plant design and construction
3. Canal System planning and designing from the rivers and storage structures
4. Environmental Hydraulics that includes the Water Supply System Design and pipe network analysis and design as well as the sewage disposal system design
5. Pipe system design for Buildings and Large Structures

Dams, Weirs and Barrages



The Bhakra Dam



HIRAKUD DAM: A MULTIPURPOSE RIVER VALLEY OF ODISHA

Dams, Weirs and Barrages

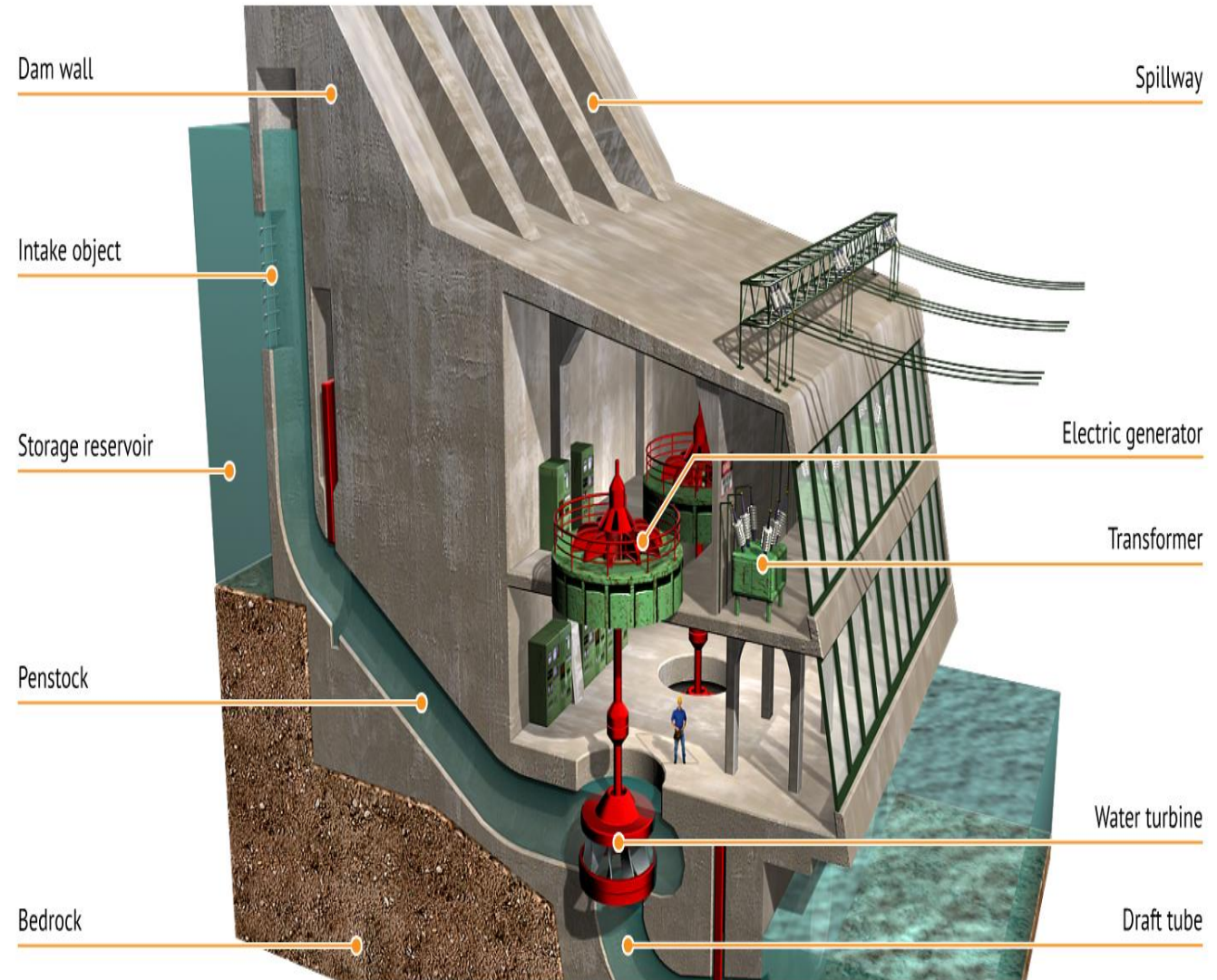
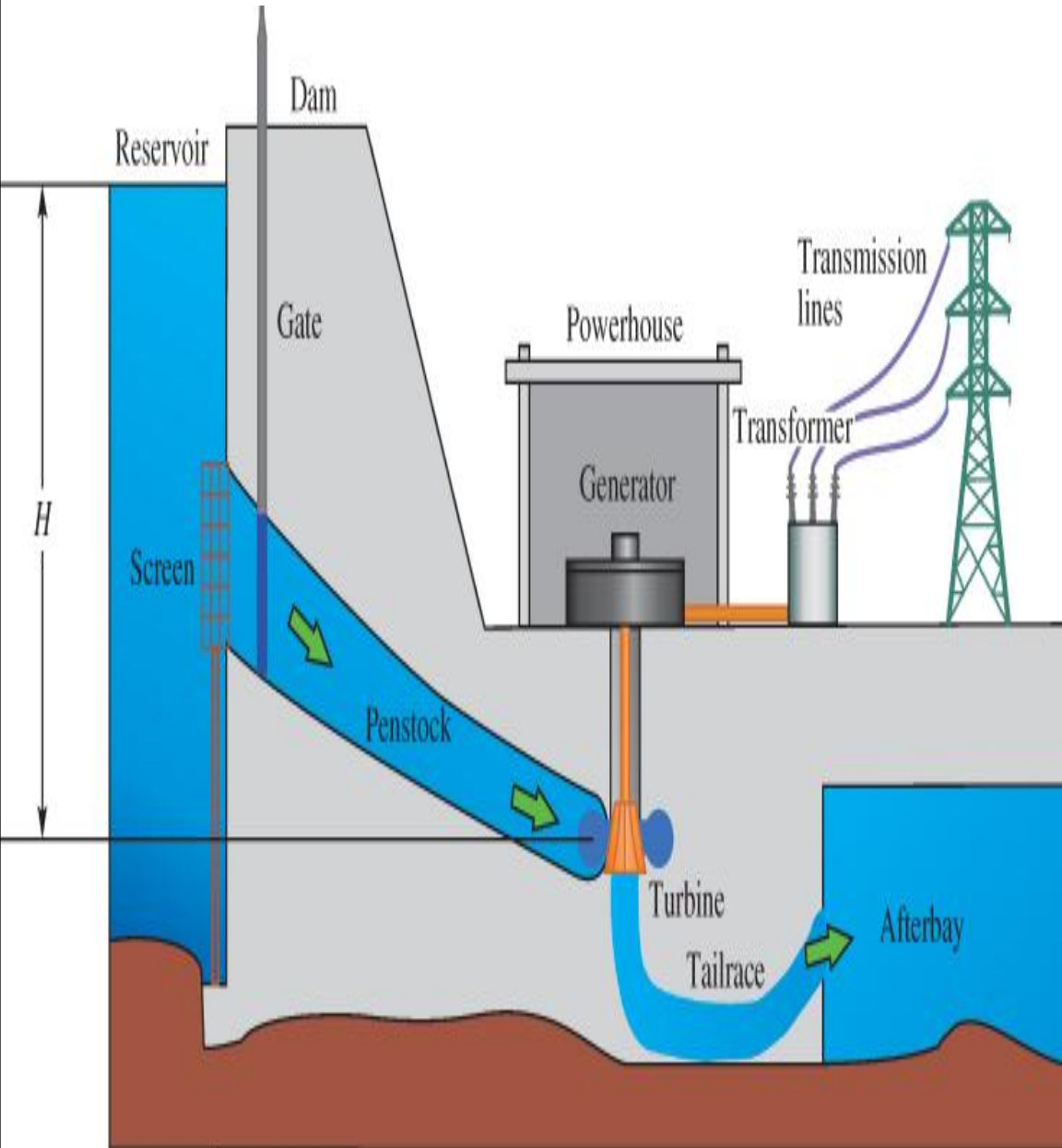


A Barrage and an a Weir

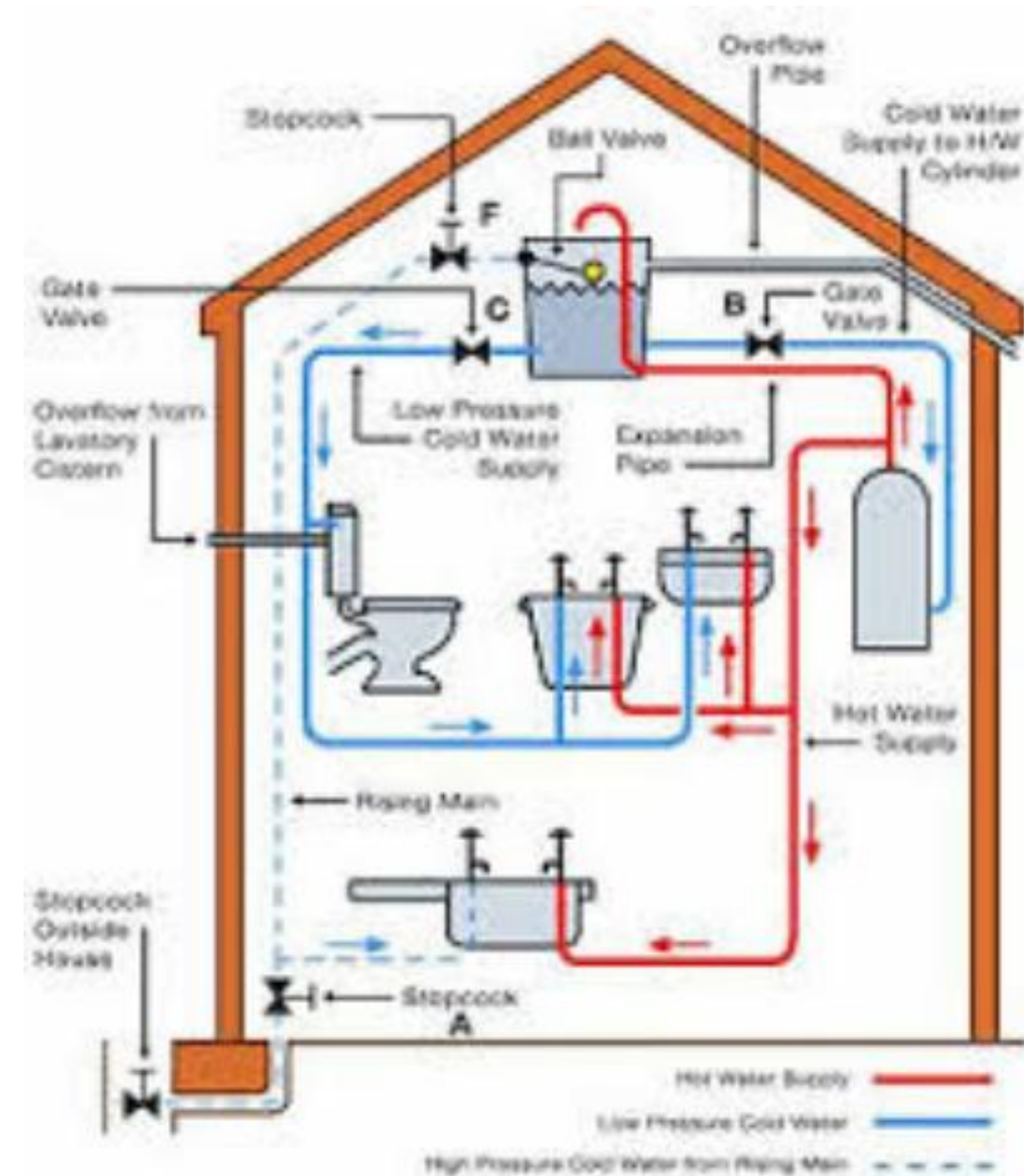
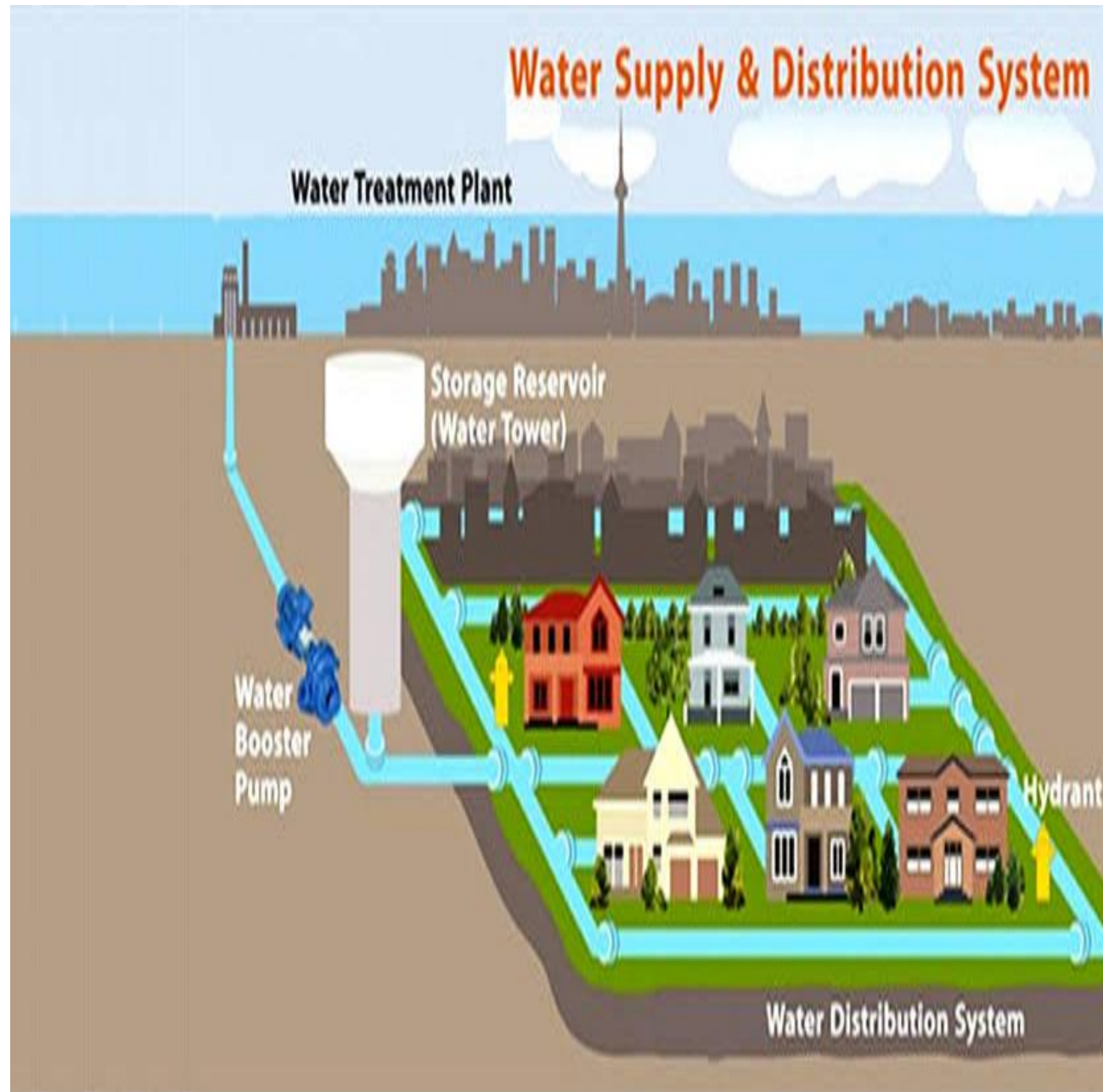
Canals and Pipe Lines



Typical Hydro-power Plants



Water Supply System



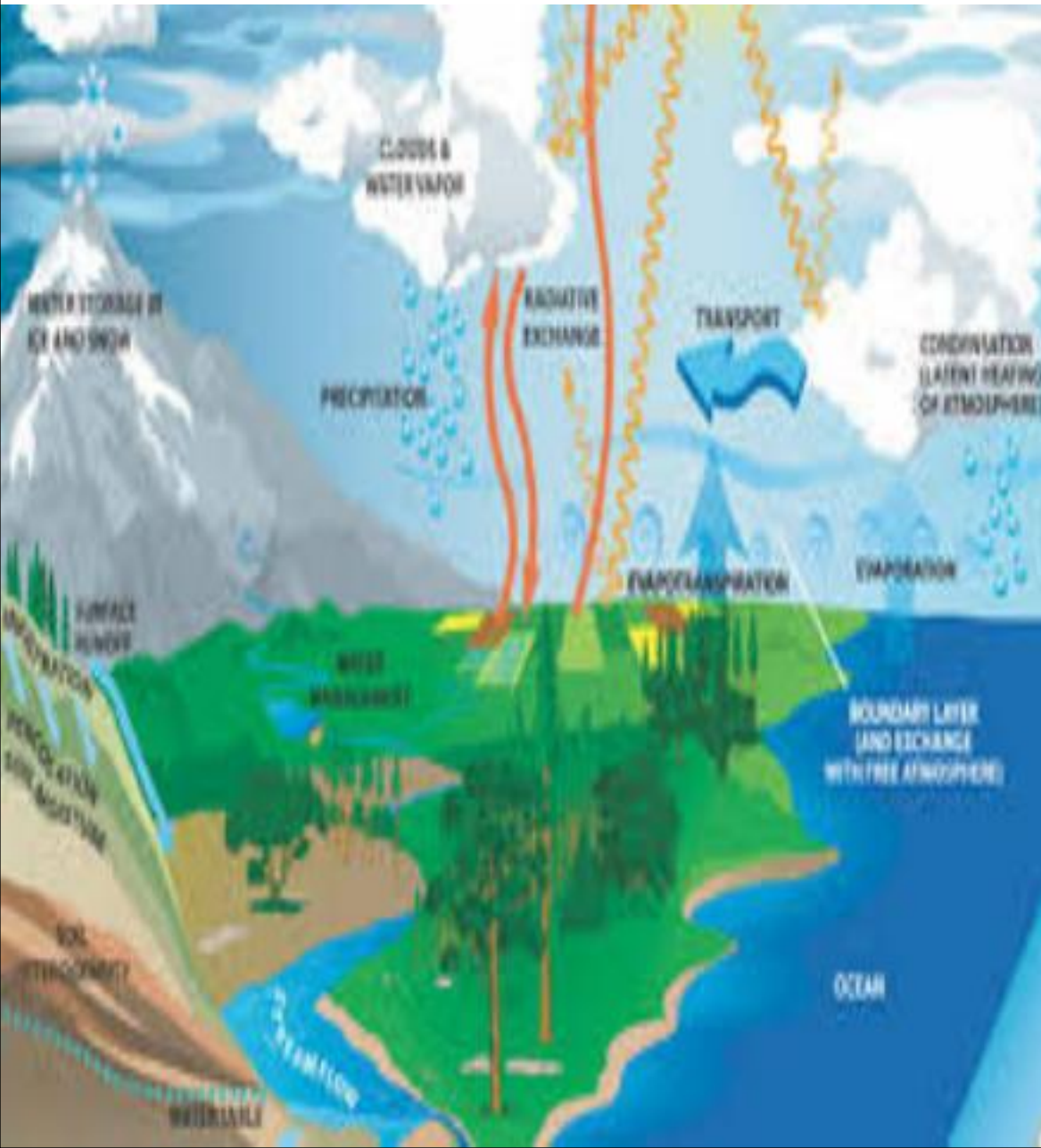
Water Resources Engineering and Management

- Water Resources Engineering: As populations grow and the availability of clean water becomes more pressing, water resource engineering addresses the planning, management, and distribution of water.

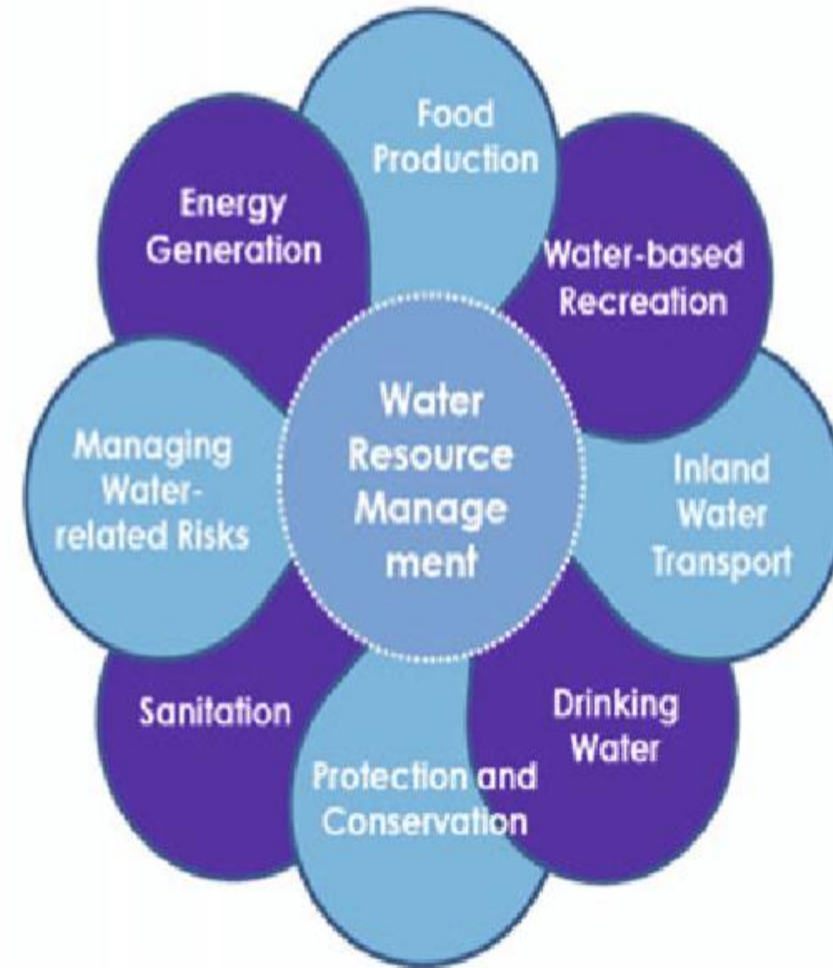
The major water consumer in this world is irrigation. Engineering and Management of supply of water in irrigation is a major discipline of water resources engineering. Thus one sub-discipline includes designing dams, reservoirs, water treatment plants, and drainage systems. 2. It often overlaps with environmental engineering, as water resource engineers also focus on minimizing ecological impacts and ensuring sustainable water usage. The Major

The fields of Hydraulics (or Fluid Mechanics) and Water Resources Engineering works in association with each other. Hydrology is also a major field of studies in water resources engineering and management. The flood management, drought management etc. are sub discipline of Hydrology. Hydrology is subdivided in to Surface Water and Ground Water studies.

Water Resources Engineering and Hydrology



WATER RESOURCE MANAGEMENT



Irrigation Engineering



ENVIRONMENTAL ENGINEERING

Though often considered as a separate engineering discipline, environmental engineering is closely linked to civil engineering. Environmental engineers tackle issues such as pollution control, waste management, and the overall environmental impact of infrastructure projects. Application of chemistry, biology and fluid mechanics in the design and operation of systems for providing drinking water, and for the treatment of municipal and industrial wastes both solids and liquids. Modelling and monitoring of the movement and behavior of pollutants in the environment. Specially in the air, water, soils and earth surface.

Based on:

- Chemistry (for understanding reactions and effects)
- Biology (for knowing the impact on living things)
- Mathematics (for modelling the responses)
- Design of waste treatment systems and Water Treatment Systems
- Study of pollution and its mitigation
- Assessment of Environmental Impact and Legal Aspects of Environmental Pollution

ENVIRONMENTAL ENGINEERING



Transportation Engineering

<https://projectinfrastructure.com/civil-engineering/>

- **Transportation Engineering:** Transportation engineers oversee the planning, design, construction and operation of transportation systems—roads, highways, railways, airports, and mass transit networks. They analyze traffic flow, optimize route networks, and implement road safety measures. With the rise of urban populations and new mobility technologies, transportation engineers are increasingly adopting data-driven approaches like intelligent transportation systems (ITS).
- **Railway Engineering:** Rail Engineers are responsible for not only the design, build, operation of the rail network but they can go on to manage the sophisticated and complex interfaces which characterize existing and new railways and infrastructure.

Road Transportation and Traffic Engineering



Railway Engineering

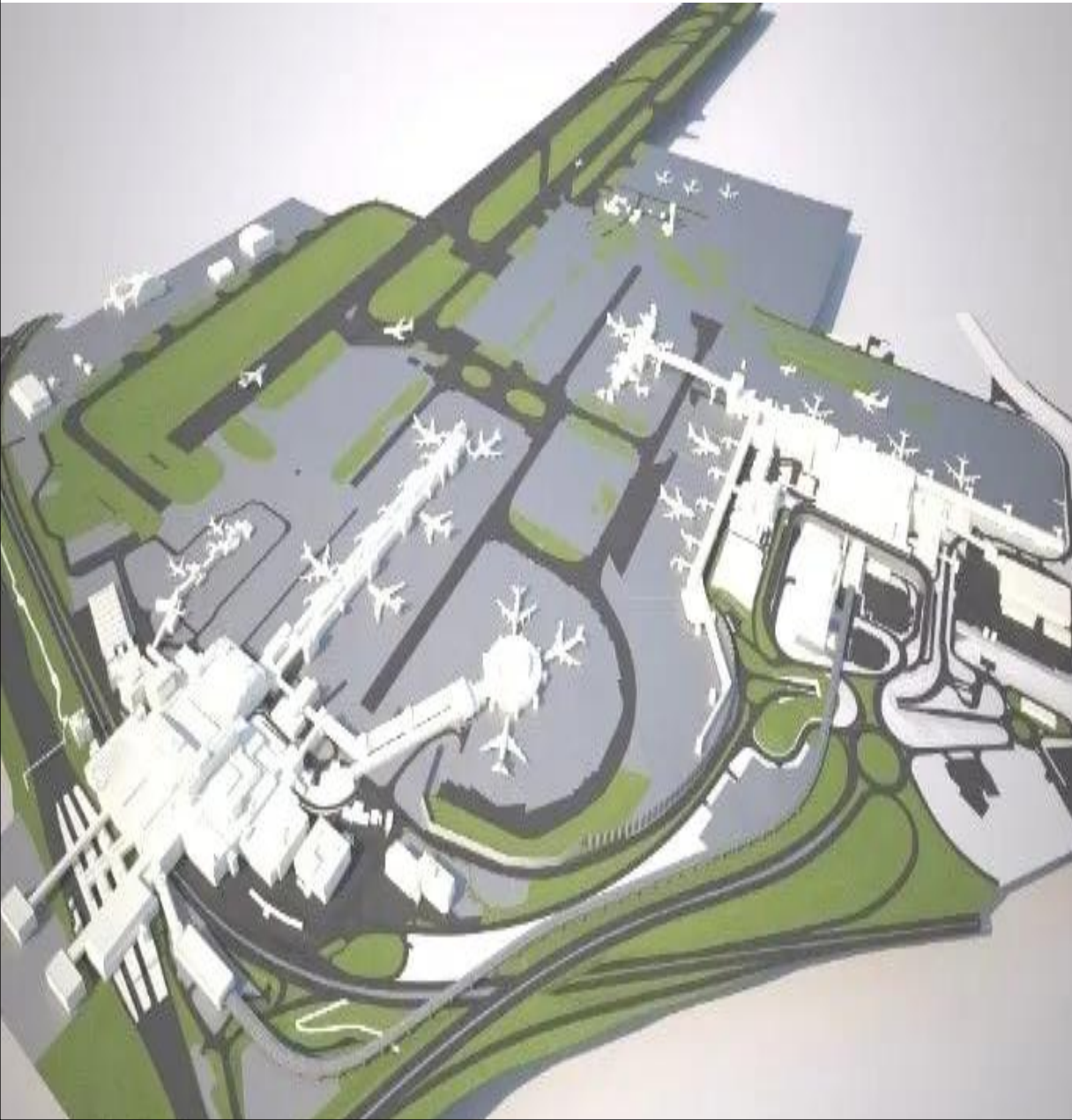


Transportation Engineering

<https://projectinfrastructure.com/civil-engineering/>

- **Aviation and Airport Civil Engineers:** These projects might involve completely new projects or changing or improving existing airports to include new technologies. These can be broken into further sub-disciplines like: Air-side infrastructure – working on runways and taxi-ways, Air-side support services – working on cargo or aircraft maintenance facilities. Terminal Construction – working within construction engineering on the design and maintenance of terminal buildings
- **Offshore Civil Engineering:** This sub-discipline of Civil Engineering deals with the design, construction and maintenance of structures intended to work in a stationary position in the ocean environment. This can be large oil-platforms that are critical to the Oil and Gas industry or for newer technologies like Offshore Wind Turbines or Tidal Energy plants. The job is not only to work on the design of these types of structures but also design permanent and temporary anchorages, and perform assessments of seabed stability. This might see an engineer undertaking conceptual or feasibility studies, on-site assessments.

Airport Engineering and Offshore Civil Engineering



Transportation Engineering

<https://projectinfrastructure.com/civil-engineering/>

- **Coastal and Marine Civil Engineering:** These types of Civil Engineering projects are increasingly important as coastal communities battle against rising sea levels and erosion and see Engineers working on sea defenses. This can see engineering roles around hard defence like barriers constructed from concrete or can see the creation of soft defence which might involve artificial or reconstructed beaches. These are some of the most complex engineering problems a structural engineer has to work on as they involve incredibly harsh environments. Engineers might also work on critical economic infrastructure like the building and maintaining ports, offshore wind farms and emerging technologies that harness tidal energy. Engineers will often have specific certifications in coastal engineering, nontechnical engineering, ports engineering, water engineering, and other fields.

Coastal, Marine, Harbour and Dock Engineering



Unit 5: Docks & Harbours

Tunnel Engineering

<https://www.gcelab.com/blog/what-is-tunnel-engineering-and-tunnel-design#What+is+Tunnel+Engineering+>

- Tunnels are passages beneath the ground that are utilized for transportation. They might be used to transport freight and passengers, as well as water and sewage. Beyond a certain depth, tunnels are more cost-effective than open cuttings. During the construction stage, tunnels normally undisturbed the surface activities like live traffic, open recreational activities, etc. Tunnels are considered to be less expensive than bridges or open steep earthwork cuttings for highway projects or transporting public utility services such as water and wastewater or gas line. The inability of these structures to be built out of natural resources such as rock and soil leads to their failure. Geological factors are expected to play a significant effect on their long-term stability. Aspects that are extremely important are such as ground and sub-ground conditions, construction sequencing, construction period, construction cost, etc. All these factors play a very important role in determining the feasibility of a tunnel project. Underground construction and tunnel engineering are genuinely interdisciplinary processes, requiring knowledge and skills from civil engineering, geological engineering, and mining engineering, as well as mechanical engineering, geophysics, electrical engineering, geology, etc.

Tunnel Engineering



Construction Engineering and Management (<https://www.ohio.edu/engineering/civil/academics>)

- According to the Journal of Architectural Engineering Technology, “Construction engineers receive training in techniques, methods, costs, scheduling, and staff management. Their top priority is to complete a job on time, on budget, and to the highest possible standard.” Construction engineers coordinate, organize and manage the day-to-day construction process, ensuring compliance with designs and plans. They can also help prepare studies about the project’s cost and timeline and collaborate with other engineers and building crews to ensure safety, efficiency and success. In addition to overseeing the project, they also communicate with key stakeholders on projects’ progress or challenges.
- Construction Management: Construction management covers the planning, coordination, and control of construction processes from start to finish. It encompasses budgeting, scheduling, quality control, and risk assessment. Construction managers ensure that projects are delivered on time, within budget, and according to the required specifications. They also manage the relationships among various stakeholders—clients, engineers, architects, contractors, and suppliers.

Construction Engineering and Management



Surveying and Geomatics <https://www.isu.edu/geomatics/the-profession/>

- Surveying is the technique of determining the relative terrestrial or three-dimensional position of different features on, above or beneath the surface of the earth by means of direct or indirect measurements. These measurement are often used to establish maps and boundaries for ownership, locations, such as building corners or the surface location of subsurface features, or other purposes required by government or civil law, such as property sales. Surveyors work with elements of geometry, trigonometry, regression analysis, physics (optics), engineering, metrology, programming languages, and the law. Surveyors do multiple tasks viz. Measure distances and angles between points on, above, and below the Earth's surface, Travel to locations and use known reference points to determine the exact location of important features, Research land records, survey records, and land titles, Look for evidence of previous boundaries to determine where boundary lines are located, Record the results of surveying and verify the accuracy of data, Prepare plots, maps, and reports, Present findings to clients and government agencies, Establish official land and water boundaries for deeds, leases, and other legal documents and testify in court regarding survey work. They provide data relevant to the shape and contour of the Earth's surface for engineering, map making, and construction projects.

Surveying and Geomatics <https://www.isu.edu/geomatics/the-profession/>

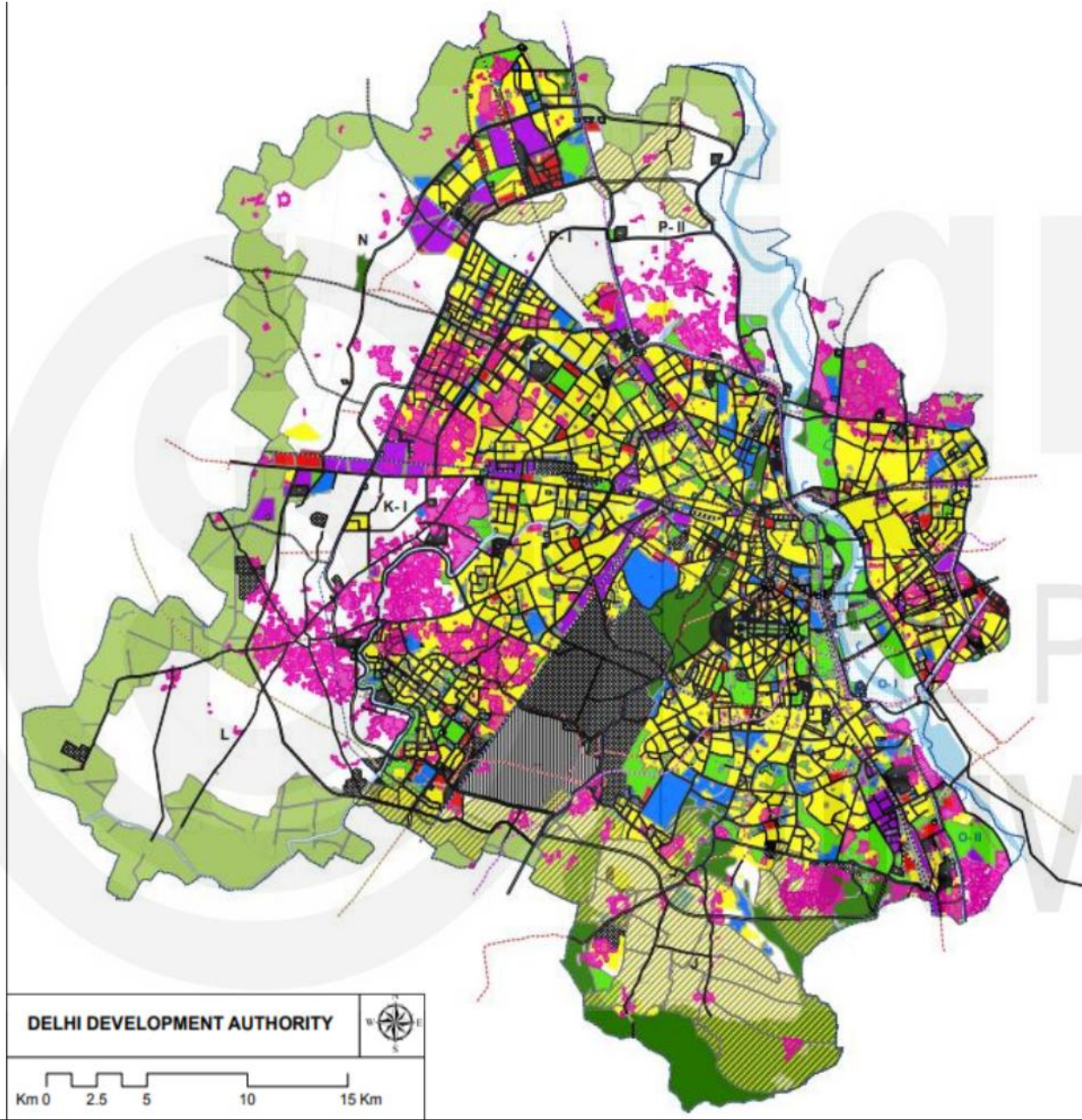
- They use a variety of measuring equipment depending upon the type of survey such as total stations, robotic total stations, theodolites, GNSS receivers, retroreflectors, 3D scanners, radios, clinometer, handheld tablets, digital levels, subsurface locators, drones, GIS, and surveying software. Surveyors then interpret and verify the results on a computer. Surveyors work with civil engineers, landscape architects, cartographers and photogrammetrists, and urban and regional planners to develop comprehensive design documents. Surveyors also work with attorneys in the areas of boundary law and legal descriptions.
- Geomatics is concerned with the collection, distribution, storage, analysis, processing, presentation of geographic data or geographic information. It consists of products, services and tools involved in the collection, integration and management of geographic data.

Surveying and Geomatics <https://www.isu.edu/geomatics/the-profession/>



Urban Planning and Development

- Urban Planning and Development: Urban planning integrates multiple aspects of civil engineering and architecture to design functional, aesthetically pleasing, and sustainable urban spaces. Urban planners work closely with civil engineers to ensure that housing, transportation, utilities, and recreational areas meet the needs of a growing population while maintaining environmental quality.
- The term —Urban Planning|| expresses the arrangements of different land uses of an urban settlement— be it a town, city, metro city or mega city in a way the settlement attains its own identity. Urban planning is a technical and political process, which is dedicated to the development of land, protection and use of the environment, public welfare, and the study of environment. Infrastructure comprising of physical and social infrastructure, such as transportation, water supply, sewerage, solid waste management, educational institutions, health institutions, post & communications, and statutory permission are very important aspects of the urban planning.



Mohenjo Daro

Nuclear or Traditional Power Civil Engineering:

- Engineers are needed for the design and build of a whole host of Civil Engineering infrastructure needed to create energy. These career paths overlap with the roles you might find in Mechanical Engineering. While it might sound like there isn't much variety in the energy sector the boom in renewal technologies mean that Civil Engineering graduates could end up working on designs for an offshore wind facility, not just the the maintenance of existing oil platforms or the decommissioning of nuclear power plants.

Nuclear or Traditional Power Civil Engineering:

